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A Novel Method for Thyroarytenoid Myofiber Culture

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Abstract

Objectives/Hypothesis: Myofiber culture has been employed to investigate muscle physiology in vitro and is well-established in the rodent hind limb. Thyroarytenoid (TA) myofiber culture has not been described, providing an opportunity to employ this method to investigate distinct TA myofiber functions. The purpose of this study was to assess the feasibility of a TA myofiber culture model.

Study Design: In vitro.

Methods: TA muscles from five Sprague Dawley rats were independently isolated and digested for 90 min. A smooth-tip, wide-bored pipette dissociated TA myofibers from cartilage, and the fibers were distributed on collagen-coated dishes and incubated at 37°C, 5% CO_2 for 2 h. Myofiber specificity was determined via immunolabeling for desmin and myosin heavy chain (MHC). Myofibers viability was assessed over 7 days via esterase assay. Additional myofibers were immunolabeled for satellite cell marker Pax-7. Glucocorticoid (GC) receptor (GR) was immunolabeled following GC treatment.

Results: The harvest technique yielded ~120 myofibers per larynx. By day 7, ~60% of the fibers remained attached and were calcein AM-positive/ethidium homodimer-negative, indicating viability. Myofibers were positive for desmin and MHC, indicating muscle specificity. Cells surrounding myofibers were positive for Pax-7, indicating the presence of myogenic satellite cells. Myofibers also responded to GC treatment as determined by GR nuclear translocation.

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Additional supporting information may be found in the online version of this article.

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Conclusion: TA myofibers remained viable in culture for at least 7 days with a predictable response to exogenous stimuli. This technique provides novel investigative opportunities regarding TA structure and function.

Level of Evidence: N/A

Keywords

desmin; glucocorticoids; larynx; myofiber; myosin heavy chain; pax7; thyroarytenoid; vocal fold; voice

INTRODUCTION

Voice disorders are the most common communication disorder across the lifespan; approximately 20 million Americans are diagnosed with a voice disorder each year[1] with total annual health care costs approaching 13 billion dollars.[2] The development of novel therapeutic strategies is a public health priority with the potential to profoundly impact millions of patients with laryngeal disease and voice-related disabilities. In vitro models involving animal-derived vocal fold (VF) myoblasts or myotubes (i.e., precursors to mature myofibers) have been employed to increase mechanistic insight regarding VF muscle health and physiology.[3–5] Although in vitro investigation offers affordability and precise experimental control, elucidation of the innate cellular, biochemical, and biomechanical interactions in organ systems is lacking, limiting translation to clinical scenarios. The development and optimization of methods to investigate intrinsic laryngeal musculature structure and function have the potential to alter and enhance clinical paradigms.

Primary myofiber culture was first described by Bekoff et al in 1977[6] and later refined by Bischoff et al in 1986[7] in rat hind limb muscles. This method involves isolating and maintaining single myofibers in culture. Myofiber culture has been employed to recapitulate the *in vivo* micro-environment due to the abundance and preservation of native myogenic satellite cells,[8–16] which are essential for muscle regeneration.[17,18] The primary application of myofiber culture, to date, has been to investigate myogenic satellite cell activation,[8,19] dissociation,[16] proliferation,[11,12,16] and differentiation.[11,16,20] Experimental conditions include myofibers subjected to growth factor treatment,[12,19] mechanical stretching,[13,19] hypercontraction,[21,22] and characterization of myofibers from dystrophic phenotypes.[23] However, this technique has not been described in the intrinsic laryngeal musculature, providing opportunities to gain mechanistic insight into unique TA physiology. The thyroarytenoid (TA) differs in fiber type composition from other skeletal muscles, potentially related to its specialized role(s) during phonation and other biological processes. Human and other animal TA muscles primarily contain type II fibers[24]; many non-laryngeal skeletal muscles (e.g., soleus) contain mostly type I fibers.[25] In the current study, we sought to establish the feasibility of rat TA primary isolated myofiber culture by assessing myofiber viability following a 7-day incubation period and immunolabeling for markers specific for muscle, basement membrane, and myogenic satellite cells immediately following isolation. Additionally, we investigated responses to pharmacological treatment frequently used in clinical settings (e.g., glucocorticoids).

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METHODS

Surgical Procedures

All experimentation was performed according to the guidelines and regulations approved by the Institutional Animal Care and Use Committee at the New York University Grossman School of Medicine. Five male Sprague–Dawley rats (400–550 g) were euthanized via inhalation of CO_2 gas. The larynx and trachea were exposed via midline incision along the sternal notch to the hyoid bone and continuously hydrated using serum-free Dulbecco's Modified Eagle Medium (DMEM). The larynx was excised (Fig. 1A) and the lumen was exposed. The entire larynx was pinned onto a Sylgard® dissection dish and submerged in serum-free DMEM at 4°C (Fig. 1B).

Mucosa Removal

An 18-gauge needle bevel was inserted into the inferior margin of the left VF directly under the mucosa just medial of the arytenoid cartilage. A sawing motion using the needle lancet was performed from the arytenoid cartilage to just lateral of the anterior commissure (Fig. 2A). This technique was repeated on the superior margin of the VF. The needle bevel was inserted distally into the inferior margin through the superior margin of the VF to separate the mucosa from the TA muscle and then inserted into the medial margin outlining the arytenoid cartilage. Fine-tipped forceps were used to grasp the mucosa at the incision medial to the arytenoid cartilage, and the mucosa was peeled from the TA using the needle to separate mucosa from the underlying muscle (Fig. 2B). The mucosa was completely removed to expose the TA muscle (Fig. 2C). The same procedure was performed on the right VF.

Arytenoid Cartilage and Thyroarytenoid Muscle Removal

Fine-tipped forceps were used to grasp the left arytenoid cartilage and a sawing motion of the needle was performed from the left arytenoid cartilage to the right arytenoid cartilage, leaving the left and right TA muscles intact at the anterior commissure (Fig. 3). The left and right arytenoid cartilages and portions of the thyroid cartilage connected to the vocalis portion of the TA were removed *en bloc*.

Digestion

The tissue was digested in 3 mL of enzyme solution (DMEM containing: 10% fetal bovine serum (FBS)), 0.3% collagenase type 3, 0.05% elastase [Worthington, USA], and antibiotic/antimycotic [Life Technologies, Grand Island, NY] for 90 min.

Fiber Trituration and Incubation

The tissue was removed from the enzyme solution and washed three times with growth medium (GM): DMEM containing 20% FBS, 10% horse serum, 2.5 ng/mL basic fibroblast growth factor; (Sigma-Aldrich, St. Louis, MO), 0.5% chick embryo extract (Accurate Chemical and Scientific, Westbury, NY), and 1% antibiotic/antimycotic solution: Thermo for 5 min. GM was replaced and the myofiber bundle attached to the arytenoid cartilage (Fig. 4A) was gently triturated using a flame-polished glass pipette tip without generating

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bubbles in the media. This trituration technique was repeated in wells containing new GM to separate bundles of fibers from connected cartilage fragments (Fig. 4B) and isolated the fibers (Fig. 4C). Trituration was repeated in wells containing new GM until fibers were completely isolated (Fig. 4D). Fragments of non-muscle tissue were removed using a 1000 μL pipette. Isolated fibers were collected and distributed on a collagen-coated dish and incubated at 37°C, 5% CO_2 for 2 h to allow non-muscle cells/tissues to adhere. Floating fibers were collected in a 15 mL conical tube supplemented with GM until the volume reached ≥10 mL or higher. Fibers were left to settle to the bottom of the tube for 5 min. The supernatant was removed slowly using a pipette until 1.5–2 mL of GM was left in the tube. The remaining medium was diluted with GM and spread onto Matrigel-coated dishes and allowed to adhere for 2–3 min. Once adhered, approximately 90% of the medium was removed. Myofibers were incubated with GM or DMEM +1% FBS to assess pharmacological treatment and myofiber survival in a condition containing lower FBS concentration without growth factors. Growth medium containing higher concentrations of FBS and growth factors may effect molecular/cellular mechanisms in vitro, potentially distorting findings in experimental treatment conditions (i.e., GCs). All samples were incubated (37°C, 5% CO_2) until experimentation.

Myofiber Viability Assay

GM or DMEM +1% FBS was replaced every 2 days and the number of adhered myofibers were counted under 50× magnification daily for 7 days. At day 7, fibers were incubated in 2 μM calcein acetoxymethyl (AM) and 4 μM ethidium homodimer in DMEM for 15 min at 37°C. In healthy cells with intact membranes, calcein–AM passively enters the cell and is hydrolyzed to calcein (green fluorescent product) by esterase enzymes; fluoresce reactivity is catalyzed and retained within the intact cell cytoplasm, indicative of cell viability.[26] Ethidium homodimer is membrane-impermeable and only reactive to dead cells with damaged membranes (red fluorescent product).[26] Fluorescence derived from calcein-AM and ethidium homodimer staining was observed using an AxioObserver microscope to quantify the survival of adhered myofibers.

Immunolabeling

Following overnight incubation with GM, a subset of myofibers was fixed in 4% paraformaldehyde-containing PBS. Fibers were treated with 1% Triton-X and incubated in PBS containing 0.05% Tween-20 and 1% bovine serum albumin. Primary and secondary antibodies are shown in Table S1. Nuclei were counterstained using 4’, 6-diamidino-2-phenylindole (DAPI; PromoCell, Heidelberg, Germany). Specimens were mounted in Fluoromount-G (Southern Biotech, Birmingham, AL).

Dexamethasone Treatment

One day following isolation, a subset of myofibers was cultured in DMEM containing 1% FBS overnight, and then treated with 1000 nM dexamethasone (DM) for 30 min. Following treatment, myofibers were washed, fixed, permeabilized, and immunolabeled for antibodies against MHC, total GC receptor (tGR), and GC receptor serine phosphorylation site 211 (s211-pGR) using identical techniques described in *Immunolabeling* section of the methodology.

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Nuclear Translocation

Quantification of tGR and s211-pGR nuclear translocation was performed using ImageJ.[27,28] Briefly, nuclear regions within each myofiber were defined by DAPI-positivity. The cytoplasm was defined as DAPI-negative and MHC-positive. Nuclear translocation was defined as the difference in staining intensity between individual nuclear and cytoplasmic regions (Δ Fluorescent intensity).

Imaging

Images were captured on a confocal laser microscope, LSM 700 (Carl Zeiss, Oberkochen, Germany).

Statistical Analyses

Nonparametric Unpaired Mann Whitney tests were employed to determine GR nuclear translocation using GraphPad Prism V.7.0 (GraphPad Software, San Diego, California, USA) due to differences in variance between experimental groups. An alpha level was set at $p = 0.05$.

RESULTS

Thyroarytenoid Myofibers Stained Positively for Desmin, Pax-7, and Laminin

On average, 122 fibers (SD = 43.7) adhered from each larynx during initial seeding. Positive immunolabeling for desmin, laminin, and Pax-7 was observed following overnight incubation. Desmin, a muscle-specific marker, was observed throughout myofibers (Fig. 5A). Pax-7, a myogenic satellite cell marker, was observed on the surface of some myofibers (Fig. 5A). Laminin, a basement membrane marker, was observed outlining myofibers (Fig. 5B).

Thyroarytenoid Fibers Remained Viable for 7 days

By 7 days, 60% of myofibers incubated in DMEM +1% FBS remained attached and viable (Table I, Fig. 6). Myofiber detachment was suppressed by GM. However, the proliferation of non-myofiber cells (e.g., fibroblasts, etc) was significant in GM, and many myotubes were formed by day 7, making discerning myofibers from myotubes difficult and myofiber count infeasible. All adhered myofibers were calcein AM-positive and ethidium homodimer-negative, indicating intact cell membranes and metabolically active myofibers.

Dexamethasone Induced GR Nuclear Translocation in TA Myofibers

Dexamethasone significantly increased tGR ($U = 0$, $p < 0.001$; Fig. 7A–C) and S211-pGR ($U = 4$, $p < 0.001$; Fig. 7D–F) nuclear translocation, consistent with previous work.[27–32]

DISCUSSION

To date, myofiber culture has been employed to investigate myogenic satellite cell behavior in various conditions including hypercontraction,[21,22] dystrophic phenotypes,[23] and growth factor treatment conditions.[12,19] Primary isolated myofiber culture is an intermediate step between *in vivo* and in vitro, providing a micro-environment similar to the *in vivo* milieu

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without the inherent shortcomings of in vitro myoblast/myotube experiments. In the current study, we sought to establish a protocol for TA primary isolated myofiber culture by confirming muscle-specific markers, assessing TA myofiber viability following a 7-day incubation period, and assessing myofiber reaction to GC treatment.

Thyroarytenoid myofibers were positively labeled for desmin, laminin, and Pax-7. Desmin is a muscle-specific intermediate filament expressed in cardiac, smooth, and skeletal muscle cells employed to determine myogenic specificity in TA myofibers. Desmin is localized to the cytoplasm of myoblasts and myotubes constituting myofibers.[33] Based on morphology and expression of desmin in the current study, isolated myofibers were determined as TA myofibers. In addition, we also observed MHC expression throughout the myofibers, further validating skeletal myofiber specificity. A similar expression of desmin was previously reported in mouse hind limb myofibers.[23]

Pax-7 is an established marker of quiescent satellite stem cells in skeletal muscle critical for skeletal muscle regeneration.[34–37] Pax-7 was observed on some of the TA myofibers, consistent with data from other myofiber models.[19–22] Previous reports localized satellite cells between the external lamina and sarcolemma of the TA myofiber.[38,39] Identifying subpopulations of satellite cells is infeasible in traditional in vitro experiments in myoblasts/myotubes. The presence of Pax-7 indicates satellite cells were retained following enzymatic digestion and isolation of TA myofibers, providing the potential to examine satellite cell proliferation, differentiation, and interaction with myofibers following various treatment/pathological conditions.

Laminin is a major component of the basement membrane in skeletal muscle and was observed surrounding TA myofibers. Although the precise location of the basement membrane is unclear in TA myofibers, laminin positivity suggests the basement membrane was partially maintained despite enzymatic isolation. Hosoyama et al.[22] reported comparable laminin distribution throughout the entire surface of intact mouse myofibers. Additionally, dispersed and disrupted laminin distribution was observed in artificially hypercontracted myofibers.[22]

Seven days following incubation in DMEM +1% FBS or GM, 60% of myofibers remained viable as determined via intracellular esterase assay, comparable to data in rat hind limb myofiber culture.[10] The 7-day endpoint was an initial step to examine the feasibility of this method in the context of acute and subacute experimental treatment paradigms. Strategies to increase viability may be considered; in the current study, culture with increased FBS concentrations yielded significant proliferation of non-myofiber cells and formation of myotubes.

The effects of DM on TA myofibers were of particular interest to assess if this model would predictably react to exogenous stimuli. Total GR and s211-pGR nuclear translocation was determined via immunolabeling as an indicator of a physiologically relevant response to DM treatment.[32,40,41] Dexamethasone significantly increased tGR and s211-pGR nuclear translocation in TA myofibers, consistent with previous work.[27–32]

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Collectively, this work confirmed myofiber specificity, viability, and predictable reactivity to clinically relevant pharmacological treatment. This model provides a foundation to investigate TA muscle-specific phenomena in an appropriate micro-environment including myogenic satellite cells and potential signaling/interactions via adherence to basement membrane components. As mentioned previously, however, muscle fiber type in the rodent TA may limit translation; the rat TA consists solely of type II fibers. The rat posterior cricoarytenoid muscle, however, contains ~10% of type 1 fibers. These issues must be considered with this model with an eye toward translation. Additionally, it is unclear if rat TA myofibers maintained type II fiber characteristics; Myofiber type is largely determined by the expression of specific MHC isoforms.[42] Future work will assess specific MHC isoforms in rat TA myofibers to determine myofiber characteristics and possible phenotype alterations during culture.

CONCLUSION

Rat TA myofibers were positive for relevant immuno-labels, remained viable for 7 days, and responded predictably to exogenous stimuli. Although isolating TA myofibers is tedious and requires magnification and precision, this model provides opportunities to explore treatment/pathological conditions in a micro-environment utilizing the advantages of both *in vivo* and in vitro settings to gain mechanistic insight into TA physiology. Additionally, primary myofiber culture has the potential to significantly reduce the number of animals required to investigate these phenomena.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

FUNDING INFORMATION

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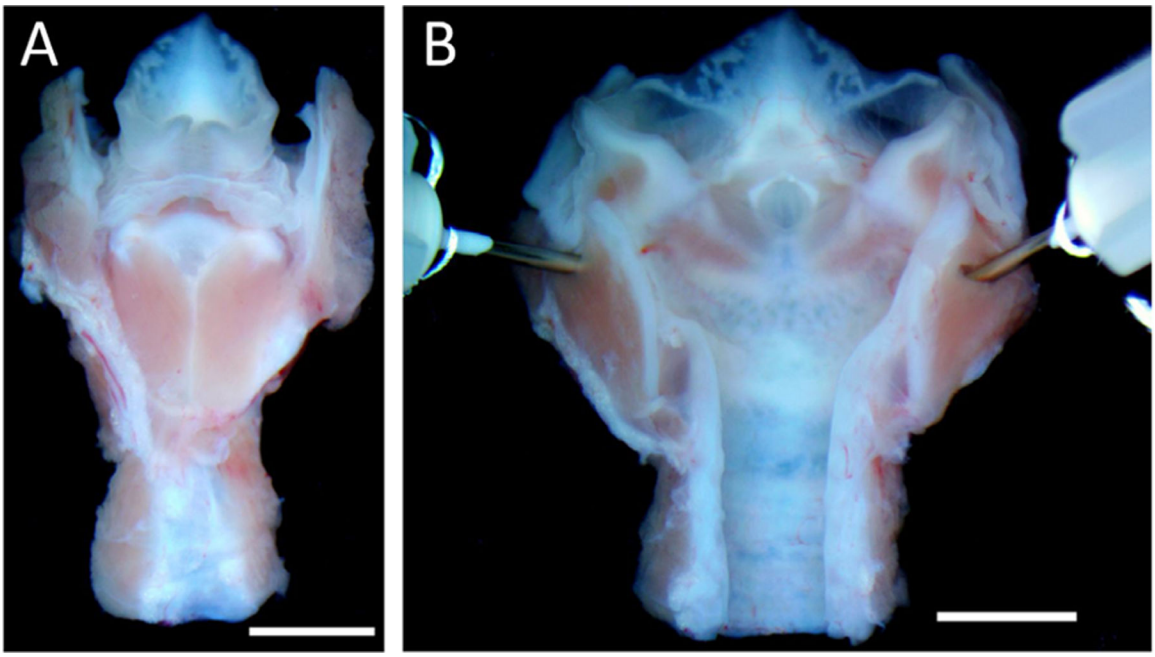


Fig. 1.
Excised rat larynx (A) and exposure of VF lumen (B). Scale bar = 2.5 mm.

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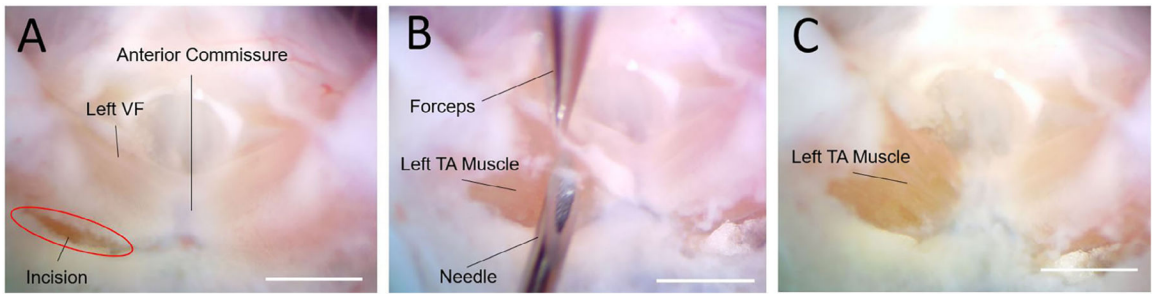


Fig. 2.

Incision from the arytenoid cartilage to just lateral of the anterior commissure (A), mucosa removal (B), and TA muscle exposure (C) in rat VF. Scale bar = 1 mm.

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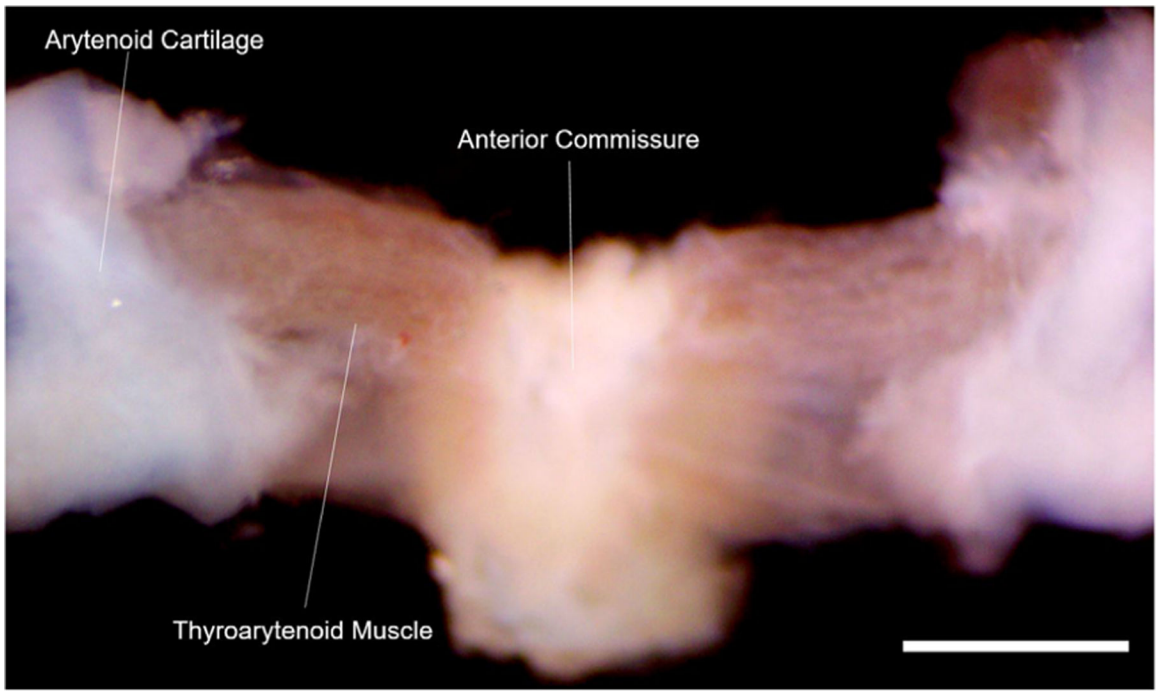


Fig. 3.
Excised rat TA muscles. Scale bar = 1 mm.

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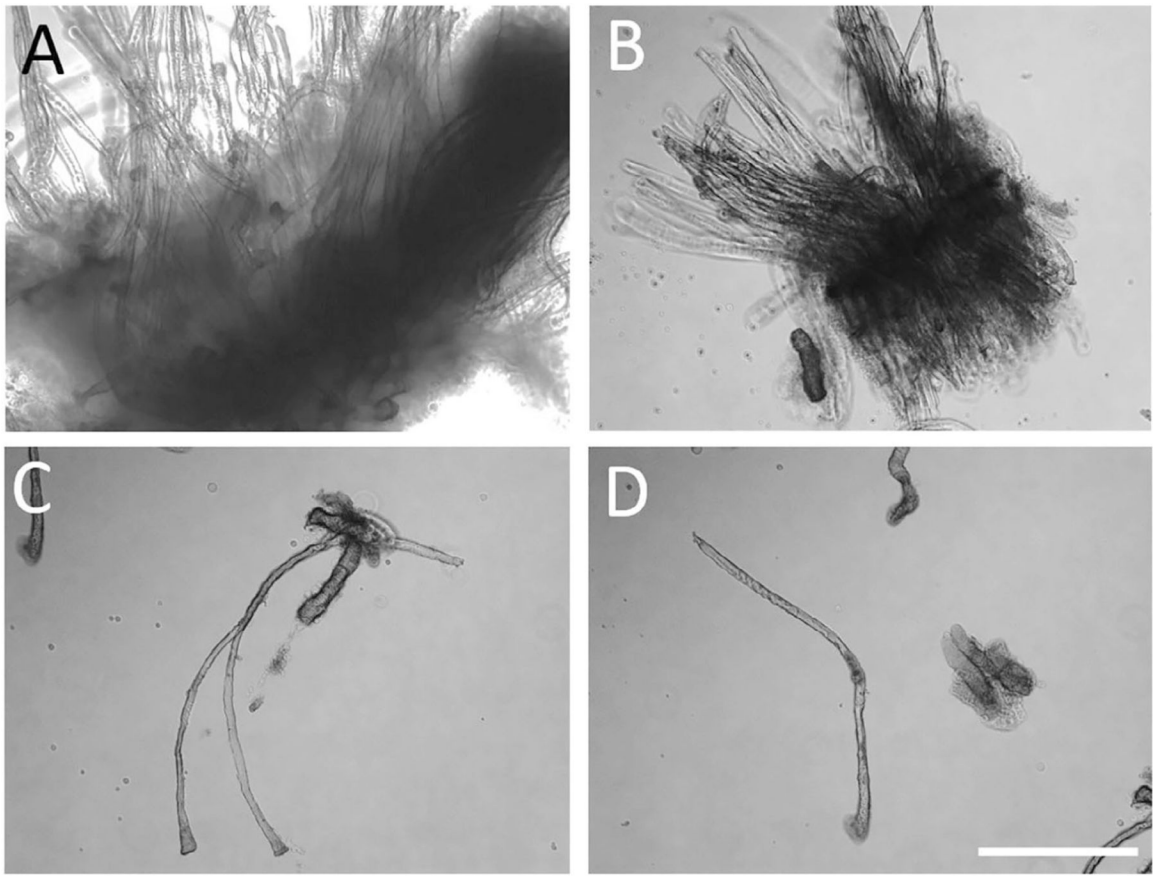


Fig. 4.

TA myofiber isolation procedure. A myofiber bundle attached to the arytenoid cartilage was incubated in GM (A). Bundles were triturated and separated from cartilage to isolate myofibers (B and C). Trituration continued until myofibers were completely isolated (D). Scale bar = 500 μm.

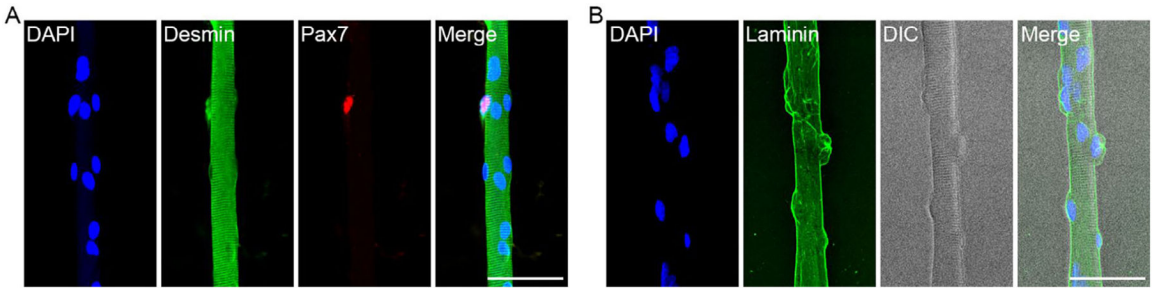


Fig. 5.

Immunolabeling for DAPI (blue), Desmin (green), Pax-7 (red) (A), and Laminin (green) (B) in rat TA myofibers. DIC = Differential Interference Contrast. Scale bar = 50 μm.

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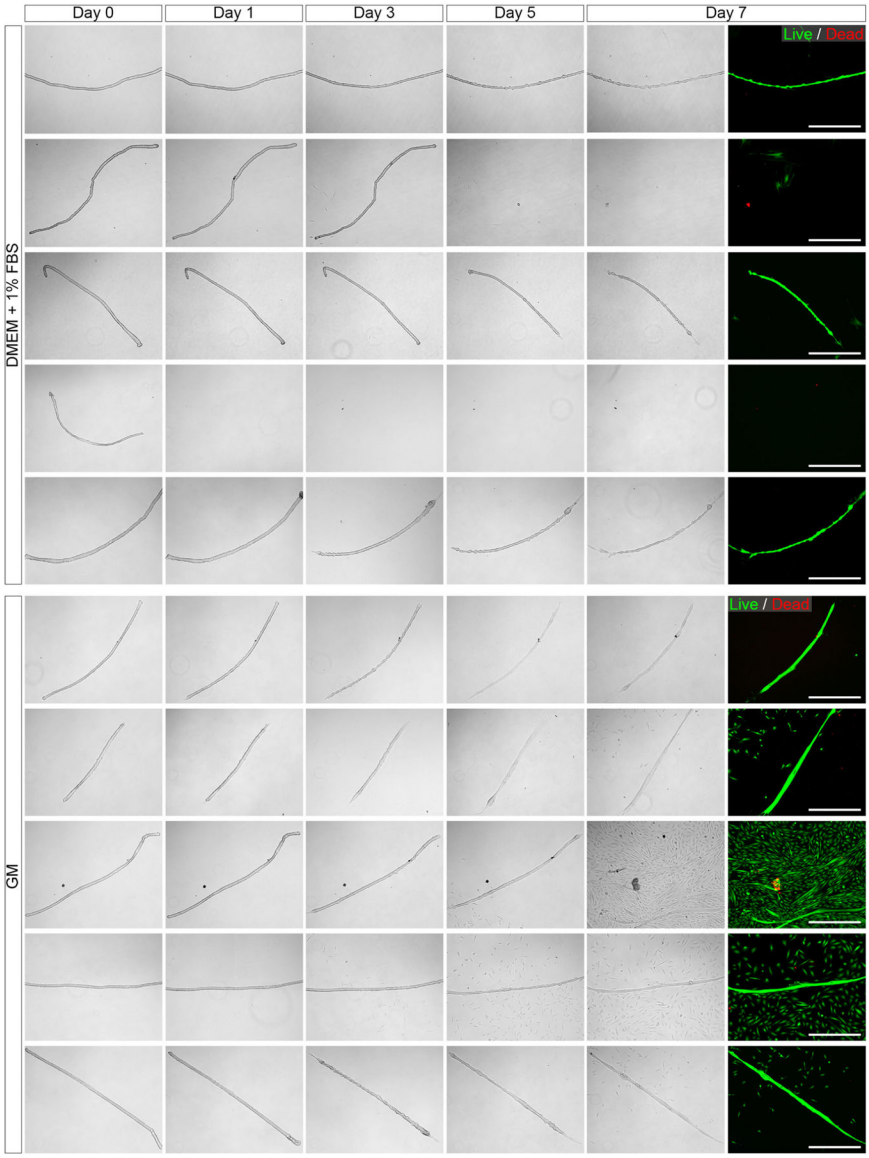


Fig. 6.

Fluorescent reactivity with esterase in rat TA myofibers intubated for 7 days in either DMEM +1% FBS or GM. Scale bar = 500 μm.

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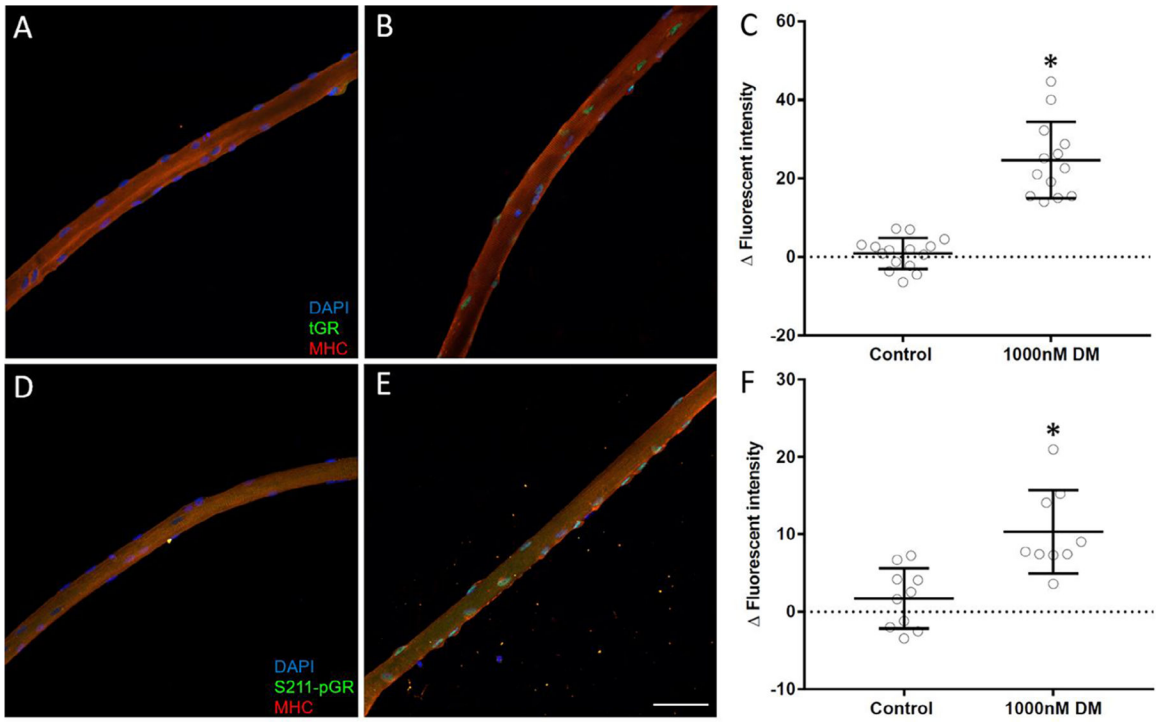


Fig. 7.

Immunolabeling for DAPI (blue), MHC (red), tGR (A and B), and s211-pGR (D and E) (green) in untreated control (A and D) and 1000 nM DM-treated (B and E) rat TA myofibers. The scatter plot represents the mean and SD of differences in fluorescent intensity between cytoplasmic and nuclear regions in control and 1000 nM DM conditions (C and F). Scale bar = 50 μm.

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TABLE I.

Average Rate of Attached Myofibers (N = 3).

Day	Average Rate of Attached Myofibers (%)
0	100
1	96
2	92
3	79
4	70
5	65
6	62
7	60